

Differentiation - Inverse Trigonometric Functions

Date _____ Period _____

Differentiate each function with respect to x .

1) $y = \cos^{-1} -5x^3$

2) $y = \sin^{-1} -2x^2$

3) $y = \tan^{-1} 2x^4$

4) $y = \csc^{-1} 4x^2$

5) $y = (\sin^{-1} 5x^2)^3$

6) $y = \sin^{-1} (3x^5 + 1)^3$

7) $y = (\cos^{-1} 4x^2)^2$

8) $y = \cos^{-1} (-2x^3 - 3)^3$